Arc spark and glow

Analytical Chemistry (University of the Punjab)

Spark and arc atomic emission spectroscopy

Spark or arc atomic emission spectroscopy is used for the analysis of metallic elements in solid samples. For non-conductive materials, the sample is ground with graphite powder to make it conductive. In traditional arc spectroscopy methods, a sample of the solid was commonly ground up and destroyed during analysis. An electric arc or spark is passed through the sample, heating it to a high temperature to excite the atoms within it. The excited analyte atoms emit light at characteristic wavelengths that can be dispersed with a monochromator and detected. In the past, the spark or arc conditions were typically not well controlled, the analysis for the elements in the sample were qualitative. However, modern spark sources with controlled discharges can be considered quantitative. Both qualitative and quantitative spark analysis are widely used for production quality control in foundry and metal casting facilities.

Arc and spark emission sources

Arc and spark are visible plasma discharges that occur between two electrodes produced when electrical current ionizes gasses in the air. Arc and spark excitation sources have been replaced with plasma or laser sources in many applications, but are still widely used in the metals industry.

Construction

Below is a sketch (figure 1) of a plasma formed between two conducting electrodes (perhaps two sharpened thoriated tungsten rods, as seen in gas tungsten arc welding). The blue region is a plasma, created by putting a sufficiently high electrical potential between the electrodes that the intervening gas is ionized. The electrodes are either metal or graphite.

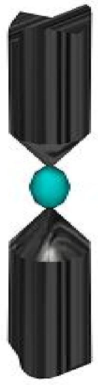


Figure 1.

If the sample to be analyzed is a metal, it can be used as one electrode. Non-conducting samples are ground with graphite powder and placed into a cup-shaped lower electrode (figure 2). Arc and spark sources can be used to excite atoms for atomic-emission spectroscopy or to ionize atoms for mass spectrometry. The arc also creates a plasma between the two electrodes. Depending on the nature of the solid material to be measured, it can either be molded into an electrode or coated onto a carbon electrode.

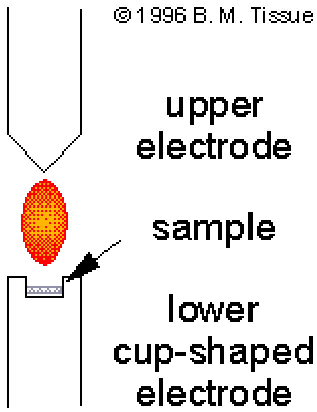


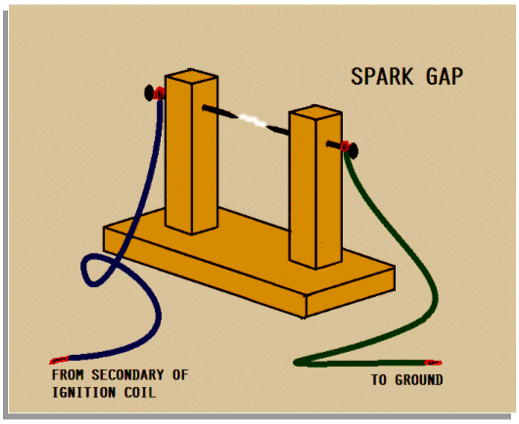
Figure 2.

Working Principle:

Arc and spark devices can be used as atomization sources for solid samples.

Spark and arc excitation sources use a current pulse (spark) or a continuous electrical discharge (arc) between two electrodes to vaporize and excite analyte atoms. A high voltage applied across a gap between two conducting electrodes causes an arc or spark to form. As the electrical arc or spark strikes the positively charged electrode, it can create a

"puff" of gas phase atoms and emission from the atoms can be measured.



https://chem.libretexts.org/Bookshelves/Analytical_Chemistry/Molecular_and_Atomic_Spectroscopy_(Wenzel)/6%3A_Atomic_Spectroscopy/6.2%3A_Atomization_Sources/6.2E%3A_Arcs_and_Sparks

Arc emission sources:

An arc, or arc discharge, is an electrical breakdown of a gas that produces a prolonged electrical discharge.

Electrical arc sources for AES use currents between 5 and 30A and burning voltages between the electrodes, for an arc gap of a few millimeters, of 20 – 60V.

D.C Arc :

Direct current arcs can be ignited by bringing the electrodes together, which causes ohmic heating. Under these conditions, d.c. arcs have a normal (decreasing) current – voltage characteristic owing to the high material ablation by thermal evaporation. The cathode glow delivers a high number of electrons carrying the greater part of the current. This electron current heats the anode to its sublimation or boiling temperature (up to 4200K in the case of graphite) and the cathode reaches temperatures of ca. 3500K as a result of atom and ion bombardment. An arc impinges on a limited number of small burning areas. The plasma temperature in the carbon arc is of the order of 6000K and it can be assumed to be in local thermal equilibrium. Arcs are mostly used for survey trace analysis requiring maximum power of detection, but they may be hampered by poor precision (relative standard deviation ≥30 %).

Alternating Current Arc:

Alternating current arcs have alternating polarity at the electrodes. Arc reignition is often by a high-frequency discharge applied at the a.c. discharge gap. Thermal effects are lower and the burning spot changes more frequently, so reproducibility is better than in d.c. arcs. Relative standard deviations of the order of 5 – 10% may be obtained.

Spark Emission sources:

A spark is an abrupt electrical discharge that occurs when a sufficiently high electric field creates an ionized, electrically conductive channel through a normally-insulating medium, often air or other gases or gas mixtures

Sparks are short capacitor discharges between at least two electrodes. The spark current is delivered by a capacitor charged to voltage by an external circuit.

- When the voltage is delivered directly by an a.c. current, an ***a.c. spark*** is produced
- . The voltage can also be delivered after transforming to a high voltage and rectifying, to produce ***d.c. spark*** discharges. In both cases, the voltage decreases within a very short time from , the burning voltage.

Formation of spark:

During the formation of a spark, a spark channel is first built up and a vapor cloud is produced at the cathode. The plasma may have a temperature of 10 000K or more, whereas in the spark channel even 40 000K may be reached. The plasma is in local thermal equilibrium, but its lifetime is short and the afterglow is soon reached. But for spark, the temperature and the background emission are low, but owing to the longer lifetime of atomic and ionic excited states, line emission may still be considerable.

Analytical Features.

- Analysis times including presparking (5 s) may be down to 10 – 20 s and the method is of great use for production control.
- Special sampling devices are used for liquid metals, e.g., from blast furnaces or convertors, and special care must be taken that samples are homogeneous and do not include cavities.
- The detection limits obtained in metal analysis differ from one matrix to another, but for many elements they are in the µg/g range .

- The power of detection of spark analysis is lower than that of d.c. arc analysis. However, by measurements in the afterglow of single sparks, or by the application of cross excitation in spark AES , there is still room for improvement of the power of detection.
- Analytical precision is high. By using a matrix line as internal standard, relative standard deviations at the 1% level and even lower are obtained.
- Matrix interference is high and relates to the matrix composition and the metallurgical structure of the sample. Therefore, a wide range of standard samples and matrix correction procedures are used. Better selection of the analytical zone of the spark is now possible with the aid of optical fibers and may decrease matrix interference.

Applications of spark emission:

- Spark emission spectrometry is a standard method for direct analysis of metallic samples and is of great use for production as well as for product control in the metals industry.
- Spark emission spectrometry enables rapid and simultaneous determination of many elements in metals, including C, S, B, P, and even gases such as N and O. Therefore, spark emission AES is complementary to X-ray spectrometry for metallurgical analyses.
- Acquisition of the spectral information from each single spark is now possible with advanced measurement systems, using integration in a small capacitor and rapid computer-controlled readout combined with storage in a multichannel analyzer.
- Spark AES has also been used for oil analysis. The oil is sampled by a rotating carbon electrode.
- Electrically insulating powders can be analyzed after mixing with graphite powder and pelleting.
- With the aid of statistical analysis, sparks with outlier signals can be rejected and the precision improves accordingly. Pulse differential height analysis enables discrimination between dissolved metals and excreted oxides for a number of elements such as Al and Ti. The first give low signals and the latter high intensities as they stem from locally high concentrations.

Reference :

Broekaert, José A. C. (2000). *Ullmann's Encyclopedia of Industrial Chemistry || Atomic Spectroscopy. , (), –.* doi:10.1002/14356007.b05_559

https://sci-hub.hkvisa.net/10.1002/14356007.b05_559

Arc and Spark Ablation

Electrical discharges of various types are often used to introduce solid samples into atomizers. The discharge interacts with the surface of a solid sample and creates a plume of a particulate and vaporized sample. The vaporization of a solid by interaction with an electrical discharge or laser beam is called ablation. The vaporized sample is then transported into the atomizer by the flow of an inert gas.

For arc or spark ablation to be successful, the sample must be electrically conducting or it must be mixed with a conductor. Ablation is normally carried out in an inert atmosphere such as an argon gas stream. Depending on the nature of the sample, the resulting analytical signal may be discrete or continuous. Several instrument manufacturers market accessories for electric arc and spark ablation and sample transport.

Note that arcs and sparks also atomize samples and excite the resulting atoms to generate emission spectra that are useful for analysis. A spark also produces a significant number of ions that can be separated and analyzed by mass spectrometry

Difference of arc and spark:

Arc	***Spark***
An arc, or arc discharge, is an electrical breakdown of a gas that produces a prolonged electrical discharge.	*A spark is an abrupt electrical discharge that occurs when a sufficiently high electric field creates an ionized, electrically conductive channel through a normally-insulating medium, often air or other gases or gas mixtures*
If the predominant behavior over the course of	*If current passes only intermittently, and most*

an experiment is to have current passing between the electrodes, with only brief (or no) interruptions, the discharge is an arc.	*of the time the gas is unionized then the discharge is a spark*
An electric arc is a continuous discharge	*While the similar electric spark discharge is momentary*
One can use arcs to carry out fractional distillation, where low-boiling elements are compounds are sampled before higher-boiling substances.	*While, sparks are more likely to sputter all elements at once than arcs.*
Free of matrix effects.	*Hardly free of matrix effects.*
Because arcs are on almost continuously, they are more nearly at thermal equilibrium than sparks.	*Not at thermal equilibrium.*
The ratio of current on-time to total experiment time is the duty cycle, If the duty cycle is more than 0.1, then its arc	*If the duty cycle is <0.1, the system is certainly a spark.*

The Glow-Discharge Technique

A glow-discharge (GD)8 device is a versatile source that performs both sample introduction and sample atomization simultaneously.

A glow discharge takes place in a low-pressure atmosphere (1 to 10 torr) of argon gas between a pair of electrodes maintained at a dc voltage of 250 to 1000 V.

The applied voltage causes the argon gas to break down into positively charged argon ions and electrons. The electric field accelerates the argon ions to the cathode surface that contains the sample. Neutral sample atoms are then ejected by a process called *sputtering*. The rate of sputtering may be as high as 100 μg/min.

The atomic vapor produced in a glow discharge consists of a mixture of atoms and ions that can be determined by atomic absorption or fluorescence or by mass spectrometry. In addition, a fraction of the atomized species present in the vapor is in an excited state. When the excited species relax to their ground states, they produce a low-intensity glow (thus, the name) that can be used for optical emission measurements.

The most important applications of the glow-discharge atomizer have been to the analysis of metals and other conducting samples, although with modification, the device has also been used with liquid samples and nonconducting materials by mixing them with a conductor such as graphite or pure copper powders. A recent advance has been the use of glow discharge emission spectroscopy for surface imaging and elemental mapping applications.

Glow-discharge sources of various kinds are available from several instrument manufacturers.